Brook Mao

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Education

University of Toronto — Toronto, Canada
Honours Bachelor of Science in Computer Science — *Expected Apr 2028*

Experience

Software Engineer Intern — May 2025 – Aug 2026
Microchip Technology Inc. — *Toronto, Canada*

- Developed a new IDE for programming Microchip FPGAs on a team of 2–5 engineers
- Extended the open-source Theia IDE with custom logic and features using object-oriented TypeScript
- Identified issues with the Theia IDE and contributed fixes to the upstream open-source project
- Added AI features to the IDE, enabling another team to integrate their specialized coding assistant
- Fixed complex bugs in the IDE to improve reliability for users, and in CI pipelines to unblock team members
- Used Claude Code to accelerate development while making sure the team understood every change
- Enhanced the UI/UX using React and CSS based on Figma designs and customer feedback
- Added remote SSH support, allowing users to connect to remote servers and program attached hardware
- Completed and maintained a hardware register viewer with read/write support, extending it to newer FPGAs
- Upgraded dependencies and refactored large portions of the codebase to keep everything compatible
- Made the IDE's Clangd C/C++ extension more robust, giving users more accurate error highlighting
- Wrote automated E2E tests with Playwright to catch regressions and cut down on manual testing
- Picked up and finished a project to make log output from multi-processor projects more readable
- Used Git, Docker, Make, Python, Bash, and Jenkins as part of the team's Linux workflow
- Created a PowerShell workflow for local development on Windows, giving team members more OS flexibility
- Reviewed dozens of pull requests, catching bugs early and pushing for better design decisions
- Authored documentation and trained customers on the product for new releases
- Tracked issues in Jira and kept project plans and progress reports in Confluence

Projects

YouTube Auto Comment | *TypeScript, Node.js, Express, PostgreSQL, GCP, Docker*

- Built a web app that monitors selected YouTube channels and posts a comment as soon as a new video drops
- Integrated Google OAuth 2.0 so the app can comment on behalf of users, with JWT cookies for sessions
- Designed a Postgres schema linking users, channels, and comments, using foreign keys to keep data consistent
- Deployed on Google Cloud Run with Docker, using Cloud Scheduler to poll YouTube for new uploads

ThreatTrace AI | *Python, Flask, TensorFlow, JavaScript, GCP*

- Collaborated in a team of 4 at Hack the Valley to develop a machine learning model that classifies cell images as healthy or malaria-infected with 95.45% accuracy, winning the Pulsar ML Challenge
- Coordinated tasks and clarified roles so the team could deliver a working prototype within 36 hours
- Built the neural network using TensorFlow and trained it on a dataset of over 27,000 images, documenting the process in a Jupyter notebook
- Preprocessed the training data and user-uploaded images to ensure model compatibility
- Deployed the model via a web app using HTML, CSS, Bootstrap, and JavaScript to create a sleek frontend, and Python with Flask to develop a REST API backend, allowing users to try the model with their own images
- Set up a CI/CD pipeline with GitHub Actions, automating Docker image builds and deployments to GCP

MultiClock | *JavaScript, MERN, Socket.IO, Tailwind CSS, NGINX, Docker, Linux, AWS, GitHub Actions*

- Built a real-time multiplayer chess clock web app using React, Node.js, Express, and MongoDB
- Used WebSockets through Socket.IO to keep every player's clock in sync instantly
- Designed a responsive UI so the app works seamlessly on both desktop and mobile
- Deployed the app on AWS using a Linux VM, configuring NGINX as a reverse proxy with HTTPS encryption
- Automated Docker image builds and AWS deployments with a GitHub Actions CI/CD pipeline

VS Code Contributions | *TypeScript, CSS*

- Contributed to Microsoft's open-source code editor, getting a small PR merged and helping improve its UI/UX

Robot Localizer | *C, LEGO EV3*

- Co-developed a LEGO EV3 robot that figures out where it is on a known map, then heads to a destination
- Implemented probabilistic localization, updating beliefs over every position and heading after each scan and move
- Improved color sensor accuracy with white balancing and averaging, classifying colors by distance in HSV space

2D Ray Tracer | *C*

- Built a 2D forward ray tracer in C that follows millions of light rays as they bounce through a scene
- Implemented ray-object intersections plus reflection, refraction, and scattering, tinting rays by material

Concurrent System Monitor | *C, Linux, Make*

- Built a Linux CLI tool that reports hardware specs and tracks CPU and memory usage in real time
- Collected each metric in its own child process, piping results to the parent and handling signals to pause or quit

Linux File Descriptor Tables | *C, Linux, Make*

- Rebuilt Linux's file descriptor tables by walking the proc filesystem, mapping each PID's FDs to paths and inodes

Mars Weather Data Pipeline | *Go, Google Cloud Storage, Google BigQuery, Power BI*

- Developed an ETL data pipeline in Go that extracts Mars weather data from the NASA API, processes it, and loads it into Google Cloud Storage and BigQuery
- Integrated Power BI with the BigQuery data warehouse to visualize Mars weather data and generate reports
- Added a Go API endpoint that triggers the pipeline automatically

Tetris | *MIPS Assembly*

- Built Tetris in MIPS assembly with a partner, using memory-mapped I/O for the bitmap display and keyboard
- Implemented all 7 tetrominoes, collision detection, line clearing, scoring, high scores, and pause/game-over screens

Toronto Asian Art Museum Mobile App | *Android Studio, Java, Firebase*

- Developed an Android app using Java and Firebase Realtime Database for managing museum artifacts
- Led a team of 6 as Scrum master, running meetings and sprint planning, and earned a 10/10 peer evaluation score
- Fixed bugs for 3 teammates so we could deliver all requirements within a 1-month deadline
- Built an interactive UI by extending AndroidX fragments into custom elements
- Applied object-oriented design patterns like Singleton and Adapter to keep the code maintainable

Portfoliify | *PHP, MySQL, Apache, JavaScript, Azure*

- Built a web app for creating and managing online profiles using PHP, MySQL, and JavaScript
- Designed a relational data model featuring 5 SQL tables to efficiently store user data
- Containerized the app with Docker and used GitHub Actions for CI/CD deployments to Azure

3D Rendering Engine | *C++, Cppcheck, Make*

- Built an object-oriented graphics program in C++ to render 3D shapes like cubes and pyramids
- Added zoom, pan, and rotation features using linear algebra
- Followed software design principles so the code stays easy to maintain and extend
- Streamlined builds with Makefiles and ran static code analysis with Cppcheck

Technical Skills

Languages: JavaScript, TypeScript, Python, C/C++, Java, SQL, Go, PHP, Bash, PowerShell, MIPS Assembly
Frameworks: React, Node.js, Express, Socket.IO, Flask, JUnit, Jest, Playwright, Bootstrap, Tailwind CSS
Developer Tools: Git, Docker, Make, GitHub Actions, Jenkins, AWS, GCP, Azure, NGINX, Firebase, MongoDB, Linux
Libraries: pandas, NumPy, Matplotlib, TensorFlow